
Evaluating Language Diffusion Models and Other Language Models with LLM Ranking Divergences

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Abstract

1 Introduction

2 Extracting LLM ranking statistics

Given a vocabulary $\mathcal{V}$, consider sequences $\bar{x} = (x_T, \dots, x_1)$ with $x_j \in \mathcal{V}$. Now consider the matrix of probabilities of an autoregressive LLM architecture $P(\bar{x})$, with P_{ij} being the probability of the j -th token at the i -th position. We will denote the vector of probabilities at the i -th location as $P_{i:}(\bar{x})$.

Given a sequence $\bar{x}$, we can define a matrix of ranked tokens:

$$R_{ij}(\bar{x}) = \text{rank}_j(P_{i:}(\bar{x})) \in \mathcal{V} \quad (1)$$

where the rank function converts the vector of probabilities in a ranked sequence of tokens and then returns the token corresponding to the j -th highest probability rank.

Given a dataset D consisting of sequences, these ranking matrices can be used to populate a histogram H , representing the LLM rank distributions under the dataset, with

$$H_j = \sum_{\bar{x} \in D} \sum_{i \in I(\bar{x})} \mathbf{1}\{R_{ij}(\bar{x}) = \bar{x}_{i+1}\}, \quad (2)$$

This histogram captures how the next-token statistics of a dataset aligns with the ranking of a given LLM.

3 Optimal transport distances between ranking histograms

We can now define an evaluation metric by comparing the ranking histogram of a real dataset H_{data} with the histogram H_{gen} obtained from an equally sized generated set from a model.

We suggest the use of an optimal transport distance:

$$d_{OT} = \text{OT}(H_{\text{data}}, H_{\text{gen}}) \quad (3)$$

This quantity vanishes when the two distributions are identical, corresponding to perfect generation under the ranking statistics, and becomes large when the distribution of the generated samples is different from the distribution of the data.

We suggest the use of a logarithmic loss:

$$c_{ij} = |\log i - \log j| \quad (4)$$

This loss emphasizes the differences in the highest ranking position while still allowing the use of rankings on a large vocabulary.

4 Closed-form optimal transport distance between ranking histograms

Let H_{data} and H_{gen} denote the ranking histograms obtained from the real dataset and from the generated samples, respectively. We assume that the two histograms have been normalized to define probability distributions over ranks,

$$h_j^{\text{data}} = \frac{H_j^{\text{data}}}{\sum_{\ell=1}^V H_{\ell}^{\text{data}}}, \quad h_j^{\text{gen}} = \frac{H_j^{\text{gen}}}{\sum_{\ell=1}^V H_{\ell}^{\text{gen}}},$$

with

$$\sum_{j=1}^V h_j^{\text{data}} = \sum_{j=1}^V h_j^{\text{gen}} = 1.$$

The optimal transport distance introduced above uses the ground cost

$$c_{ij} = |\log i - \log j|.$$

This is a one-dimensional transport problem on the ordered points

$$z_j = \log j, \quad j = 1, \dots, V.$$

Therefore, the optimal transport distance admits a closed-form expression as the one-dimensional Wasserstein-1 distance between the two empirical distributions on the logarithmic rank axis.

Define the cumulative discrepancy up to rank k as

$$\Delta_k = \sum_{j=1}^k (h_j^{\text{data}} - h_j^{\text{gen}}).$$

Then the optimal transport distance is

$$d_{\text{OT}} = \sum_{k=1}^{V-1} |\Delta_k| (\log(k+1) - \log k).$$

This expression follows from the fact that, in one dimension, the amount of mass that must cross the boundary between consecutive ranks k and $k+1$ is exactly the cumulative excess mass Δ_k . Moving this mass across the boundary incurs cost $z_{k+1} - z_k = \log(k+1) - \log k$. Summing the absolute transported mass across all adjacent boundaries gives the closed-form optimal transport distance above.

If the histograms are not normalized but have the same total mass, the same formula can be applied directly to the counts. If the total masses differ, the histograms should first be normalized, or an unbalanced optimal transport distance should be used instead.

Algorithm 1 LLM Rank-Wasserstein Evaluation

Require: Real dataset $\mathcal{D}_{\text{data}}$, generated dataset $\mathcal{D}_{\text{gen}}$

Require: Reference autoregressive LLM p_θ

Require: Vocabulary size V

Ensure: Log-rank optimal transport distance d_{OT}

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1: function RANKHISTOGRAM( $\mathcal{D}, p_\theta$ )
2:   Initialize histogram  $H \leftarrow \mathbf{0} \in \mathbb{R}^V$ 
3:   for all sequences  $\bar{x} = (x_1, \dots, x_T) \in \mathcal{D}$  do
4:     for  $i = 1$  to  $T - 1$  do
5:       Compute next-token probabilities
          
$$p_\theta(\cdot \mid x_1, \dots, x_i)$$

6:       Let  $r_i$  be the rank of the observed next token  $x_{i+1}$  under this distribution
          
$$r_i = \text{rank}_{p_\theta(\cdot \mid x_1, \dots, x_i)}(x_{i+1})$$

7:       Update histogram
          
$$H_{r_i} \leftarrow H_{r_i} + 1$$

8:     end for
9:   end for
10:  Normalize histogram
          
$$h \leftarrow \frac{H}{\sum_{j=1}^V H_j}$$

11:  return  $h$ 
12: end function

13: Compute real-data rank histogram
          
$$h_{\text{data}} \leftarrow \text{RANKHISTOGRAM}(\mathcal{D}_{\text{data}}, p_\theta)$$

14: Compute generated-data rank histogram
          
$$h_{\text{gen}} \leftarrow \text{RANKHISTOGRAM}(\mathcal{D}_{\text{gen}}, p_\theta)$$

15: Initialize cumulative discrepancy  $\Delta_0 \leftarrow 0$ 
16: Initialize distance  $d_{\text{OT}} \leftarrow 0$ 
17: for  $k = 1$  to  $V - 1$  do
18:   Update cumulative discrepancy
          
$$\Delta_k \leftarrow \Delta_{k-1} + h_{\text{data},k} - h_{\text{gen},k}$$

19:   Add transport cost across boundary  $k \leftrightarrow k + 1$ 
          
$$d_{\text{OT}} \leftarrow d_{\text{OT}} + |\Delta_k| (\log(k + 1) - \log k)$$

20: end for
21: return  $d_{\text{OT}}$ 
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